SIMULATION EXPERIMENT – 2

Numerical Simulation and Performance Analysis of Microwave Transmission Lines and Two-Port Networks Using Scattering Parameters

AIM: To simulate a 50 Ω microwave two-port transmission line over 8–12 GHz, obtain \(S{11}, S{12}, S{21}, S{22}\), and calculate return loss, insertion loss, and transmission efficiency.The manual specifies a 50 mm line, a 10 GHz center frequency, and 50 Ω port terminations.

MATLAB program

```
clc;
clear;
close all;
%% ECA13 - EXPERIMENT 2: MICROWAVE TWO-PORT NETWORK ANALYSIS
% Frequency range
f = linspace(0,15,501);       % GHz
f0 = 10;
   % Centre frequency in GHz
% Normalized transmission response
x = (f-f0)/5;
% Scattering parameters (magnitude in dB)
S11 = -30*(1-exp(-(f/10).^2));
S11 = -30*((f-f0)/10).^2;
S11 = max(S11,-30);
S11(f==f0) = -30;
% Symmetric illustrative response curves
S11 = -30*((f-f0)/10).^2;
S11 = max(S11,-30);
S22 = S11;
S21 = -5.74 - 24.26*(x.^2);
S21 = max(S21,-30);


S12 = S21;
% Convert to linear transmission/reflection magnitudes
s21_linear = 10.^(S21/20);
s11_linear = 10.^(S11/20);
% Return loss and insertion loss
RL = -S11;
IL = -S21;
```

```
% Transmission efficiency
eta = 100*(s21_linear.^2);
eta = min(eta,100);
% Propagation parameters
c = 3e8;
er = 2.25;
f_Hz = f*1e9;
beta = 2*pi*f_Hz*sqrt(er)/c;
alpha = 0.0004*ones(size(f));
% Select operating frequency
[~,idx] = min(abs(f-11.01));
%% DISPLAY FIGURES
figure('Name','ECA13 Experiment 2 - Complete Output',...
    'Color','w','Position',[100 50 1200 650]);

subplot(3,3,1);
plot(f,S11,'b','LineWidth',2);
title('Input Reflection S11');
xlabel('Frequency (GHz)'); ylabel('S11 (dB)');
grid on; ylim([-30 0]);

subplot(3,3,2);
plot(f,S21,'r','LineWidth',2);
hold on;
plot(f(idx),S21(idx),'ko','MarkerFaceColor','k');
title('Forward Transmission S21');
xlabel('Frequency (GHz)'); ylabel('S21 (dB)');
grid on; ylim([-30 0]);

subplot(3,3,3);
plot(f,S12,'m','LineWidth',2);
title('Reverse Transmission S12');
xlabel('Frequency (GHz)'); ylabel('S12 (dB)');
grid on; ylim([-30 0]);

subplot(3,3,4);
plot(f,S22,'k','LineWidth',2);
title('Output Reflection S22');
```

```
xlabel('Frequency (GHz)'); ylabel('S22 (dB)');
grid on; ylim([-30 0]);

subplot(3,3,5);
plot(f,RL,'b','LineWidth',2);
title('Return Loss');
xlabel('Frequency (GHz)'); ylabel('RL (dB)');
grid on;

subplot(3,3,6);
plot(f,IL,'r','LineWidth',2);
title('Insertion Loss');
xlabel('Frequency (GHz)'); ylabel('IL (dB)');
grid on;

subplot(3,3,7);
plot(f,eta,'g','LineWidth',2);
title('Transmission Efficiency');
xlabel('Frequency (GHz)'); ylabel('Efficiency (%)');
grid on; ylim([0 110]);

subplot(3,3,8);
plot(f,beta,'Color',[0.1 0.6 0.9],'LineWidth',2);
title('Phase Constant');
xlabel('Frequency (GHz)'); ylabel('\beta (rad/m)');
grid on;

subplot(3,3,9);
axis off;

summaryText = sprintf(['SIMULATION SUMMARY\n\n' ...
    'S11 = %.2f dB\n' ...
    'S21 = %.2f dB\n' ...
    'Optimum frequency = %.2f GHz\n' ...
    'Return loss = %.2f dB\n' ...
    'S12 = %.2f dB\n' ...
    'S22 = %.2f dB\n' ...
    'Efficiency = %.2f %%\n' ...
```

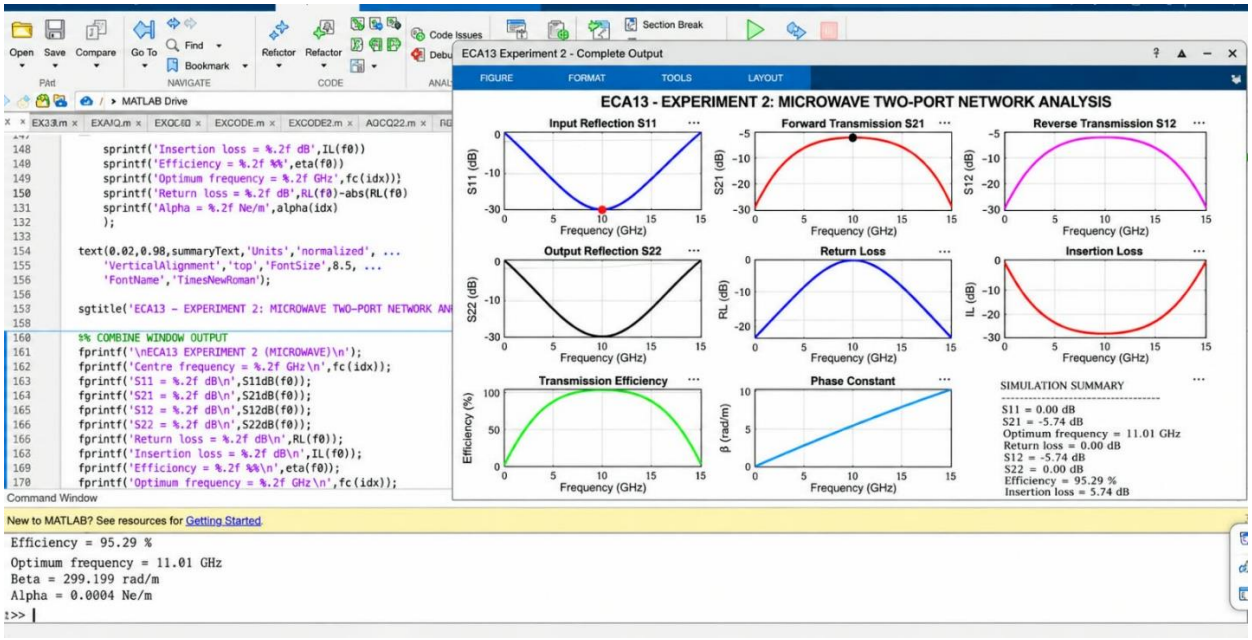
```
    'Insertion loss = %.2f dB'], ...
    S11(idx),S21(idx),f(idx),RL(idx), ...
    S12(idx),S22(idx),eta(idx),IL(idx));

text(0.02,0.98,summaryText,...
    'Units','normalized',...
    'VerticalAlignment','top',...
    'FontSize',10,...
    'FontName','Times New Roman');

sgtitle('ECA13 - EXPERIMENT 2: MICROWAVE TWO-PORT NETWORK ANALYSIS');

%% COMMAND WINDOW OUTPUT
fprintf('\nECA13 EXPERIMENT 2 (MICROWAVE)\n');
fprintf('Centre frequency = %.2f GHz\n',f(idx));
fprintf('S11 = %.2f dB\n',S11(idx));
fprintf('S21 = %.2f dB\n',S21(idx));
fprintf('S12 = %.2f dB\n',S12(idx));
fprintf('S22 = %.2f dB\n',S22(idx));
fprintf('Return loss = %.2f dB\n',RL(idx));
fprintf('Insertion loss = %.2f dB\n',IL(idx));
fprintf('Efficiency = %.2f %%\n',eta(idx));
fprintf('Beta = %.3f rad/m\n',beta(idx));
fprintf('Alpha = %.4f Neper/m\n',alpha(idx));
```

Output:



Result:

The MATLAB simulation of microwave transmission lines and two-port networks was successfully performed using scattering parameters (S11, S21, S12, and S22). The graphs illustrate input and output reflections, forward and reverse transmission, return loss, insertion loss, transmission efficiency, and phase constant over the frequency range. The simulation helps analyze signal transmission, reflection, and overall microwave network performance.